

Causal Phase Geometry Research Program

Minimal Draft, Competing Interpretations of the Phase Layer,
and a Program of Experimental Tests

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Abstract

A working hypothesis is proposed that combines the common core of several earlier constructions: causal microstructure, local proper time, discrete $U(1)$ holonomies, optical observables, and thermodynamic coarse-graining. The fundamental kinematics is specified by a locally finite causal structure with phase transports on oriented links. In an appropriate continuum limit, causal order and volume information should determine a Lorentzian metric, while the phase transports should determine a $U(1)$ connection. Light is treated as an excitation of the connection and as a null probe of the full geometry, rather than as the material of space or as a stretched string.

Two competing interpretations of the phase layer are fixed. Under hypothesis H_R , the layer is a physical five-dimensional geometry and, after reduction, contains a dynamical radion. Under hypothesis H_P , the circle is an internal phase space over each point of the four-dimensional base; there is no free radion and no physical KK tower. The minimal limit of H_P coincides with the Einstein-Maxwell system, and any possible new physics can reside only in a correlation functional generated by the joint coarse-graining of causal and phase variables.

The idea of a limiting geometry $g_{\mu\nu}^*$ with a homogeneous coasting regime $a_*(t) \propto t$ is also considered, but is not treated as an established fact. The observed metric may be a nonequilibrium deformation of this fixed point, while “synchronization” may mean a decrease of relative entropy toward a stationary state. Such a decomposition becomes physical only after an independent derivation of the relation between the state of the network and the metric deformation.

The main purpose of the text is not to select a correction term in advance, but to define a program for extracting constraints from cosmological data, galaxy rotation curves and baryonic profiles, local tests of scalar interactions, and the open FAIR-MAST archive. The result should be either a set of common response parameters suitable for constructing a mathematical model, or a quantitative refutation of both branches.

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1 Purpose and Status of the Construction

This text is not a completed theory. Its purpose is to fix the minimal mathematical scheme common to two earlier drafts and to separate it from interpretations for which no derivation currently exists.

Three levels of claims are used.

Established mathematical structures:

1. locally finite causal sets;
2. Lorentzian geometry and null cones;
3. principal $U(1)$ bundles, connections, and holonomies;
4. geometric optics in curved spacetime;
5. the Einstein–Maxwell system;
6. the local thermodynamic interpretation of the Einstein equation in Jacobson’s approach.

Working hypothesis:

$$\boxed{(\mathcal{C}, \prec, U_{xy}) \xrightarrow{\text{coarse-graining}} (\mathcal{M}_4, g_{\mu\nu}, A_\mu)} \quad (1)$$

The causal structure and its volume information generate the metric sector, while the $U(1)$ transports generate the phase connection. Both sectors are

treated as large-scale manifestations of one microscopic structure, but they are not identified with one another.

The present draft does not assume:

- that the phase circle is a physical fifth spatial dimension;
- that a free radion field exists;
- that null radiation has the equation of state of a stretched string;
- that the expansion of the Universe must be linear;
- that redshift has already been explained without expansion;
- that the Tully-Fisher relation or the radial acceleration relation has already been derived from the microdynamics;
- that the cosmological constant is a demonstrated contribution of phase geometry.

2 Discrete Causal Kinematics

2.1 Causal order

The fundamental structure is a locally finite partially ordered set

$$\mathcal{C} = (X, \prec), \quad (2)$$

where $x \prec y$ means that event x causally precedes event y .

For causally related events, define the interval

$$I(x, y) = \{z \in X \mid x \prec z \prec y\}. \quad (3)$$

The number of its elements serves as a discrete analogue of four-dimensional volume:

$$V_4(x, y) \sim v_* |I(x, y)|. \quad (4)$$

A timelike distance can be estimated by the length of a maximal chain,

$$L(x, y) = \max_{\gamma: x \rightsquigarrow y} |\gamma|, \quad \tau(x, y) \sim \tau_* L(x, y). \quad (5)$$

Equations (4) and (5) are asymptotic correspondences expected only in a manifold-like regime. They do not define the dynamics of the causal structure.

2.2 Causality and metric scale

Causal order alone determines the light cones and the conformal class of the metric, but not the absolute scale of proper times and volumes. Schematically,

$$\prec \implies [g_{\mu\nu}], \quad (6)$$

where metrics related by

$$g_{\mu\nu} \sim \Omega^2(x) g_{\mu\nu} \quad (7)$$

have the same null structure. Additional volume information is needed to reconstruct the full metric:

$$\boxed{\text{causal order} + \text{volume measure} \longrightarrow \text{full Lorentzian geometry}} \quad (8)$$

in an appropriate continuum limit.

This does not imply the rule

$$\text{more elements} \implies \text{slower time}. \quad (9)$$

The density of discretization, interval volume, and the rate of a particular clock are distinct concepts. Any relation among them must be derived from dynamics rather than introduced by verbal identification.

3 Local Time

A spacetime point does not possess its own clock. Proper time is defined along a timelike worldline γ :

$$\boxed{\tau[\gamma] = \frac{1}{c} \int_{\gamma} \sqrt{-g_{\mu\nu} dx^{\mu} dx^{\nu}}} \quad (10)$$

Thus, the statement “time runs more slowly in this region” means a comparison of proper times for two specified families of observers. It is not a local scalar independent of the chosen worldlines and the comparison procedure.

In a weak stationary field,

$$ds^2 \simeq - \left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) d\mathbf{x}^2, \quad (11)$$

and for stationary clocks,

$$d\tau \simeq \left(1 + \frac{\Phi}{c^2}\right) dt. \quad (12)$$

Within this limited regime, free fall can be described intuitively as motion toward smaller accumulated proper time of stationary observers. Full gravitation, however, also includes spatial curvature, shift, rotation of reference frames, and dynamical degrees of freedom of the metric.

4 Limiting Geometry and Entropic Relaxation

A recurring intuition in the earlier drafts is the existence of a limiting, or “ideal,” geometry. To avoid introducing a second physical Universe, this idea should be interpreted as a hypothesis about a fixed point of coarse-graining:

$$g_{\mu\nu}^* = \text{stationary large-scale regime of the causal-phase network.} \quad (13)$$

In the homogeneous sector, one may investigate the ansatz

$$a_*(t) = \frac{t}{t_*}, \quad H_*(t) = \frac{1}{t}, \quad \ddot{a}_* = 0, \quad (14)$$

but it is not identified with the established history of the observed Universe. It is useful to decompose the physical metric as

$$\boxed{g_{\mu\nu} = \Omega^2(x) (g_{\mu\nu}^* + q_{\mu\nu})}, \quad (15)$$

where Ω sets the volume and conformal scale, while $q_{\mu\nu}$ is a nonconformal, tidal, and anisotropic deformation. A pure factor Ω preserves unparametrized null trajectories; therefore lensing and Weyl curvature generally require nonzero $q_{\mu\nu}$.

For the homogeneous limit,

$$a(t) = a_*(t)\Omega(t), \quad (16)$$

we obtain

$$H(t) = \frac{1}{t} + \frac{\dot{\Omega}}{\Omega}, \quad \frac{\ddot{a}}{a} = \frac{\ddot{\Omega}}{\Omega} + \frac{2}{t} \frac{\dot{\Omega}}{\Omega}. \quad (17)$$

Hence the regime $a_* \propto t$ does not generate acceleration by itself; the entire observed difference must reside in the dynamics of Ω and $q_{\mu\nu}$.

The notion of “entropic synchronization” should be formulated not as a decrease in total entropy, but as a decrease in statistical distance from the fixed point. For a microscopic-state distribution ρ_t and a stationary distribution ρ_* , define

$$\boxed{\mathcal{D}(t) = D_{\text{KL}}(\rho_t || \rho_*), \quad \dot{\mathcal{D}} \leq 0,} \quad (18)$$

while still allowing

$$\dot{S}_{\text{total}} \geq 0. \quad (19)$$

At present, only the structural relation

$$\ln \Omega = \Phi[\mathcal{D}, \nabla \mathcal{D}, \dots], \quad q_{\mu\nu} = \mathcal{Q}_{\mu\nu}[\rho_t, U_t], \quad \Phi(0) = 0, \quad (20)$$

is admissible, without selecting the functions Φ and $\mathcal{Q}_{\mu\nu}$. Any history $a(t)$ can be written formally as $t\Omega(t)$; therefore (15) becomes predictive only when Ω and $q_{\mu\nu}$ are independently derived from the state of the causal-phase network.

5 $U(1)$ Phase Geometry on a Causal Structure

Associate to each oriented causal link $x \prec y$ an element

$$U_{xy} \in U(1). \quad (21)$$

For a charged field,

$$\psi_x \longrightarrow g_x \psi_x, \quad g_x \in U(1), \quad (22)$$

the transport transforms as

$$U_{xy} \longrightarrow g_x U_{xy} g_y^{-1}. \quad (23)$$

For a causal chain

$$\gamma : x = x_0 \prec x_1 \prec \dots \prec x_n = y, \quad (24)$$

define

$$U(\gamma) = \prod_{j=0}^{n-1} U_{x_j x_{j+1}}. \quad (25)$$

Because there are no oriented causal cycles, discrete curvature may be defined by comparing two distinct paths with the same endpoints:

$$\boxed{W(\gamma_1, \gamma_2) = U(\gamma_1)U(\gamma_2)^{-1}}. \quad (26)$$

In the continuum limit, one assumes

$$U_{xy} \simeq \exp \left(iq \int_x^y A_\mu dx^\mu \right), \quad (27)$$

and the relative holonomy should reproduce the curvature

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (28)$$

In this picture, the photon is neither the phase of an individual particle nor a one-dimensional string under tension, but a propagating excitation of the gauge connection.

6 Continuum Limit and Geometric Representation

The minimal continuum limit has the structure of a principal bundle,

$$U(1) \hookrightarrow \mathcal{P} \xrightarrow{\pi} \mathcal{M}_4, \quad (29)$$

where $g_{\mu\nu}$ is the metric on the base and A_μ is a local representation of the connection on the $U(1)$ layer.

For a unified geometric notation, one may use an auxiliary Kaluza–Klein-type metric:

$$\boxed{\mathrm{d}\hat{s}^2 = g_{\mu\nu} \mathrm{d}x^\mu \mathrm{d}x^\nu + \ell_*^2 (\mathrm{d}\theta + \kappa A_\mu \mathrm{d}x^\mu)^2}. \quad (30)$$

In the minimal version,

$$\theta = \text{internal phase coordinate}, \quad \ell_* = \text{fixed normalization of the layer}. \quad (31)$$

Thus, (30) is used as a geometric packaging of $g_{\mu\nu}$ and A_μ , not as a claim about a physical fifth direction. In particular, the core contains neither

$$\ell_* \rightarrow b(x), \quad (\nabla b)^2, \quad \text{a tower of physical KK modes}. \quad (32)$$

A literal five-dimensional theory with a dynamical radion remains a separate possible extension, but it is not part of the present minimal draft.

7 Two Competing Interpretations of the Phase Layer

For the experimental program, two ontologically distinct branches and the standard null model must be separated in advance.

7.1 Hypothesis H_R : physical radion

In this branch, the five-dimensional representation is interpreted literally, and the radius of the layer becomes a field,

$$b(x) = b_0 \exp\left(\frac{\phi(x)}{M_\phi}\right). \quad (33)$$

The low-energy functional has the general structure

$$\Gamma_R = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} (\nabla\phi)^2 - V(\phi) - \frac{1}{4} B_F(\phi) F_{\mu\nu} F^{\mu\nu} + \mathcal{L}_m(g, \phi, \Psi) \right]. \quad (34)$$

A free radion implies an additional scalar pole in the response function. In the linear approximation, the characteristic susceptibility has the form

$$\chi_R(\omega, k) \propto \frac{1}{m_\phi^2 + c_\phi^2 k^2 - \omega^2 - i\gamma\omega}. \quad (35)$$

Consequently, the same branch must manifest itself consistently, or be constrained, through a fifth force, equivalence-principle violation, variations of α and G , scalar polarization of gravitational waves, laboratory response to strong electromagnetic fields, and the cosmological evolution of ϕ .

7.2 Hypothesis H_P : internal phase space

In this branch,

$$U(1) \hookrightarrow \mathcal{P} \rightarrow \mathcal{M}_4 \quad (36)$$

is an internal bundle rather than a physical spatial direction. One fixes

$$\delta\ell_* = 0, \quad (37)$$

and there are no independent states $b(x)$ and $\phi(x)$. The effective functional is written as

$$\Gamma_P = \Gamma_0[g, A, \Psi] + \Gamma_{\text{corr}}[g, A, \Psi]. \quad (38)$$

The correlation term may be nonlocal and collective, but in the pure H_P version it must not generate a universal freely propagating scalar pole. If such a pole appears after coarse-graining, it is experimentally a physical scalar degree of freedom regardless of the original interpretation.

7.3 Null model H_0

The null model includes the standard system of general relativity, electrodynamics, and the accepted astrophysical inventory:

$$H_0 : \quad \Gamma = \Gamma_0, \quad (39)$$

with dark matter and a cosmological constant, or with another prespecified standard cosmological baseline. Any claimed anomaly must be compared not only between H_R and H_P , but also with H_0 , including systematic errors and conventional plasma or galactic astrophysics.

Table 1: Qualitative differences between the two branches.

Feature	H_R : physical radion	H_P : phase bundle
Independent scalar mode	Yes, unless fully stabilized	No in the minimal version
Fifth force and WEP	Arise naturally and must be suppressed or detected	Not required
Variations of α and G	Possible and correlated with ϕ	Do not follow from a fixed layer
Local response to strong F^2	Possible through $B_F(\phi)$	Only through ordinary ED or Γ_{corr}
Cosmological dynamics	May proceed through $\phi(t)$ and $V(\phi)$	Only through a collective geometric response
KK tower	Possible in a literal five-dimensional realization	Absent

8 Light as a Null Probe of the Full Geometry

In geometric optics, the wave covector of light satisfies

$$g^{\mu\nu} k_\mu k_\nu = 0, \quad (40)$$

and rays follow null geodesics:

$$k^\nu \nabla_\nu k^\mu = 0. \quad (41)$$

The local speed of light equals c for every freely falling observer. This does not imply that the coordinate speed is constant in every coordinate system, and it does not define a proper time for the photon:

$$d\tau_{\text{null}} = 0. \quad (42)$$

The frequency measured by an observer with four-velocity u^μ is

$$\omega = -u_\mu k^\mu. \quad (43)$$

Thus, the general redshift has the invariant form

$$1 + z = \frac{(-u_\mu k^\mu)_{\text{em}}}{(-u_\mu k^\mu)_{\text{obs}}}. \quad (44)$$

Equation (44) unifies Doppler, gravitational, and cosmological contributions. It shows that the observed shift is a comparison of null transport with the local temporal directions of the source and the observer.

Gravitational lensing is not interpreted as an artificial elongation of the path intended to “synchronize” light with slow clocks. Clock rates and ray focusing are distinct consequences of the same metric.

For a light bundle, its transverse geometry obeys the optical equations. In particular, the angular-diameter distance satisfies the schematic equation

$$\frac{d^2 D_A}{d\lambda^2} = - \left(\frac{1}{2} R_{\mu\nu} k^\mu k^\nu + |\sigma|^2 \right) D_A, \quad (45)$$

where σ is the shear of the null congruence. In a metric theory with photon-number conservation, the reciprocity relation also holds:

$$D_L = (1 + z)^2 D_A. \quad (46)$$

Therefore, any alternative cosmological interpretation must simultaneously reproduce z , D_A , D_L , lensing, and redshift drift, rather than only one luminosity-redshift diagram.

9 Minimal Effective Dynamics

The most conservative continuum functional is

$$\Gamma_0[g, A, \Psi] = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} (R - 2\Lambda) - \frac{1}{4e^2} F_{\mu\nu} F^{\mu\nu} + \mathcal{L}_m(g, A, \Psi) \right]. \quad (47)$$

This functional already contains bidirectional interaction:

- the metric determines the propagation and energy of the electromagnetic field;
- the electromagnetic stress-energy tensor curves the metric;
- charged matter is a source of the $U(1)$ connection.

The mere fact of a common geometric origin does not require a new force or an additional cosmology. New physics is possible only if joint coarse-graining generates a nonzero residual functional,

$$\Gamma_{\text{eff}} = \Gamma_0 + \Gamma_{\text{corr}}[g, A, \Psi]. \quad (48)$$

Its geometric response can be defined as

$$H_{\mu\nu}^{\text{corr}} = -\frac{2}{\sqrt{-g}} \frac{\delta\Gamma_{\text{corr}}}{\delta g^{\mu\nu}}, \quad (49)$$

and the correction to the connection equation through

$$\mathcal{J}_{\text{corr}}^\nu = \frac{1}{\sqrt{-g}} \frac{\delta\Gamma_{\text{corr}}}{\delta A_\nu}. \quad (50)$$

The field system then takes the symbolic form

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G (T_{\mu\nu}^{\text{m}} + T_{\mu\nu}^{\text{EM}}) - H_{\mu\nu}^{\text{corr}}, \quad (51)$$

$$\nabla_\mu F^{\mu\nu} = e^2 J^\nu - e^2 \mathcal{J}_{\text{corr}}^\nu, \quad (52)$$

with coefficients depending on the chosen normalization.

Diffeomorphism and gauge invariance impose consistency conditions on (49) and (50). Therefore, Γ_{corr} cannot be chosen arbitrarily to fit observations.

The form of Γ_{corr} is not specified in this text. Computing it from the discrete model is the central unresolved task.

10 Thermodynamic Interpretation

Local horizon thermodynamics may be applied to the physical four-dimensional geometry. In equilibrium form one uses

$$\delta Q = T_U \delta S_H, \quad S_H = \frac{k_B c^3}{4\hbar G} A, \quad (53)$$

together with the null Raychaudhuri equation. Under suitable conditions, this leads to the Einstein equation as a local equation of state.

In the minimal phase version, the $U(1)$ layer is not a physical spatial dimension. Therefore, one cannot automatically use a five-dimensional horizon area or a five-dimensional Jacobson equation. Maxwell's equations must follow from variation of the phase connection, while the electromagnetic stress-energy tensor enters the heat flux and the geometric equation.

The scheme is

thermodynamics of the causal base $\rightarrow$ gravitational equation, variation of the phase connection $\rightarrow$ Maxwell equation, $T_{\mu\nu}^{\text{EM}} \rightarrow$ backreaction on the metric.	(54)
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Positive entropy production by itself determines neither the sign of the pressure nor $w = -1$.

11 Cosmological Status

For a homogeneous and isotropic metric,

$$ds^2 = -c^2 dt^2 + a^2(t) d\Sigma_k^2, \quad (55)$$

the minimal functional (47) gives standard Einstein-Maxwell cosmology with a cosmological constant. An isotropically averaged electromagnetic field has

$$p_{\text{EM}} = \frac{1}{3} \rho_{\text{EM}}, \quad \rho_{\text{EM}} \propto a^{-4}, \quad (56)$$

and does not behave as a cosmological constant.

A homogeneous change in the coefficient of dt^2 without a change in spatial geometry is a redefinition of the time coordinate and does not by itself generate an observable cosmological redshift. A physical alternative to expansion must specify the full metric and calculate observables from null geodesics.

Expansion does not follow logically from the single fact that $z > 0$. An alternative model must, however, reproduce the entire related set

$\{z, D_L(z), D_A(z), \dot{z}, \text{time-profile dilation, lensing, BAO, CMB}\}.$	(57)
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In particular, one cannot substitute $k = k(t)$ into the standard Friedmann equations while leaving the FLRW metric unchanged. If the spatial curvature is dynamical, a more general geometry is required, for which the field and optical equations must be derived anew.

A possible contribution from Γ_{corr} may take a vacuum-like form on a homogeneous background,

$$H_{\mu\nu}^{\text{corr}} \simeq -\rho_{\text{corr}} g_{\mu\nu}, \quad (58)$$

but this is a target of calculation, not a postulate. The present draft does not claim that phase geometry has already explained the cosmological constant.

12 Galactic Status

The original Tully–Fisher relation uses luminosity as an observational proxy. Its baryonic form relates the asymptotic velocity to the total baryonic mass:

$$M_b \propto V_f^4. \quad (59)$$

The radial acceleration relation imposes a more local condition,

$$g_{\text{obs}}(r) = \mathcal{F}(g_{\text{bar}}(r)). \quad (60)$$

These relations do not establish the primacy of luminosity, acceleration, or the electromagnetic field. In the present hypothesis, they are treated as possible boundary conditions on Γ_{corr} .

In the weak-field regime,

$$ds^2 = -(1 + 2\Psi/c^2)c^2 dt^2 + (1 - 2\Phi/c^2)d\mathbf{x}^2. \quad (61)$$

The dynamics of slow bodies primarily measures Ψ , whereas gravitational lensing depends on the combination $\Phi + \Psi$. Therefore, a successful model must simultaneously reproduce

$$V_{\text{rot}}(r), \quad \alpha_{\text{lens}}(r), \quad \text{cluster dynamics and structure growth.} \quad (62)$$

Neither luminosity nor the internal charge content of neutral baryonic matter by itself yields a derivation of (59). A calculation of the collective response of the common causal-phase structure is required.

13 Preliminary Guidance from SPARC Analyses

Previous computational experiments with the SPARC database are not regarded as evidence for the hypothesis. They are used to formulate cleaner statistical tests. The database itself contains homogeneous photometric profiles and rotation curves for disk galaxies [12].

At least two different parameters were used in earlier scripts and must be kept strictly separate.

The local fraction of the dynamical discrepancy:

$$\mu_{\text{dyn}}(r) = 1 - \frac{V_{\text{bar}}^2(r)}{V_{\text{obs}}^2(r)} = 1 - \frac{g_{\text{bar}}(r)}{g_{\text{obs}}(r)}. \quad (63)$$

The gas-weighted global indicator:

$$\mu_{\text{gas}} = \frac{\int (g_{\text{gas}}/g_{\text{bar}}) g_{\text{obs}} \mathrm{d}r}{\int g_{\text{obs}} \mathrm{d}r}. \quad (64)$$

Neither should currently be called a fundamental coupling constant. μ_{dyn} algebraically contains the observed velocity, while μ_{gas} simultaneously uses the gas fraction and a dynamical weight.

Preliminary plots showed the following persistent motifs:

1. the relative dynamical discrepancy increases as the local surface brightness and baryonic acceleration decrease;
2. gas-dominated radial points, after pooling different galaxies, form a mean curve that rises toward a plateau; however, universal parameters for this curve have large scatter and were estimated from the same data;
3. the absolute residual velocity and the relative discrepancy behave differently, because

$$V_{\text{res}} = \sqrt{V_{\text{obs}}^2 - V_{\text{bar}}^2} = V_{\text{obs}} \sqrt{\mu_{\text{dyn}}}; \quad (65)$$

4. the apparent dependence of the residual plateau on gas fraction is partly built into the definitions of both axes and is strongly confounded with total mass, size, and surface density.

The most informative new formulation separates predictor and response across different regions of a galaxy. An outer baryonic indicator can be defined without V_{obs} :

$$X_{i,\text{out}} = \left\langle \frac{g_{\text{gas}}}{g_{\text{bar}}} \right\rangle_{r > R_{\text{cut}}}, \quad (66)$$

while the inner response can be defined as a residual relative to a nonparametric mean relation:

$$Y_{i,\text{in}} = \langle \log g_{\text{obs}} - \hat{m}(\log g_{\text{bar}}) \rangle_{r < R_{\text{eff}}}. \quad (67)$$

The test

$$Y_{i,\text{in}} = \beta X_{i,\text{out}} + \boldsymbol{\gamma}^T \mathbf{C}_i + \epsilon_i \quad (68)$$

must control for M_{bar} , V_f , disk size, surface brightness, inclination, distance, mass-to-light ratio, data quality, and an angular-momentum proxy. A robust $\beta \neq 0$ after these checks would indicate a nonlocal statistical relation, but would not yet prove a causal-phase origin.

A more general object is the radial response kernel,

$$\Delta_i(r) = \int K(r, r') X_i(r') \mathrm{d}r' + \epsilon_i(r). \quad (69)$$

Local dynamics corresponds to a kernel concentrated near $r = r'$. Robust off-diagonal elements $K(r, r')$, after controlling for ordinary astrophysics, may provide empirical guidance for a nonlocal Γ_{corr} .

14 Elements Retained from Earlier Hypotheses

The following are retained from the causal-thermodynamic draft:

- the primacy of causal order;
- discrete $U(1)$ holonomies;
- separation of the base metric from the phase connection;
- the possibility of an auxiliary KK representation;
- the thermodynamic interpretation of the gravitational equation;
- the requirement that cosmology be derived from the complete effective tensor rather than from analogies.

The following are retained from the causal null-network draft:

- emphasis on null structure as a primary part of geometry;
- the need to distinguish light cones from the metric rate of local clocks;
- the idea of testing cosmology through light propagation;
- the conjecture that galactic correlations may reflect a collective geometric response to the baryonic distribution.

15 Elements Excluded from the Core

The following claims from the earlier texts are not included in the new hypothesis.

1. **Null radiation is not identified with a stretched string.** Isotropic averaging of null dust gives $w = +1/3$, whereas $w = -1/3$ applies to a network of objects with longitudinal tension.
2. **Isotropization does not create a time axis.** The condition $\langle n^i \rangle = 0$ does not eliminate the second moment $\langle n^i n^j \rangle$ and does not transform a spatial distribution into a purely temporal substrate.
3. **$E \propto N^2$ is not postulated.** The square of a sum of amplitudes in a coherent process is not a universal law for the total energy of a network.

4. **Black holes are not treated as an external sink of the Universe's energy.** Their mass does not disappear from the exterior geometry; an open subsystem requires an explicit boundary and a definition of quasilocal energy.
5. $R = ct$ **is not postulated as the observed history.** Only the investigation of $a_* \propto t$ as a possible fixed point is admitted. The physical history must follow from Ω , $q_{\mu\nu}$, and observational tests.
6. **A dynamical $k(t)$ is not inserted into unchanged FLRW equations.** Such a replacement changes the class of metrics.
7. **A free radion is not assumed.** In the minimal version, the phase layer has a fixed internal normalization.
8. **Dark matter and dark energy are not derived.** The corresponding effects may be discussed only after Γ_{corr} has been calculated.

16 Minimal Discrete Dynamics

For the $U(1)$ sector, a natural first candidate is a Wilson-like functional on causal diamonds:

$$S_{U(1)}[U; \mathcal{C}] = \frac{1}{g_*^2} \sum_D w_D [1 - \text{Re } W_D]. \quad (70)$$

For a small phase,

$$W_D = e^{i\Phi_D}, \quad 1 - \text{Re } W_D \simeq \frac{1}{2} \Phi_D^2. \quad (71)$$

The required continuum limit is

$$S_{U(1)} \longrightarrow -\frac{1}{4e^2} \int \mathbf{d}^4x \sqrt{-g} F_{\mu\nu} F^{\mu\nu}. \quad (72)$$

The gravitational part must assign a probabilistic weight to causal structures and possess a continuum limit containing the Einstein-Hilbert action. The full scheme is

$$Z = \sum_{\mathcal{C}} \int \mathcal{D}U \mathcal{D}\Psi \exp \left[\frac{i}{\hbar} S[\mathcal{C}, U, \Psi] \right]. \quad (73)$$

The functional Γ_{corr} must emerge from (73) after coarse-graining. Until this step is carried out, there is no basis for selecting its form by hand.

17 Development and Experimental-Test Program

The purpose of the program is not to fit a separate function to each class of data, but to reconstruct common response functions and test whether a single parameter set exists for H_R or H_P . The analysis must always include the null model H_0 .

17.1 Stage 0: formalization of competing models

Before any new statistical analysis, minimal parameterizations must be fixed.

For the physical radion,

$$\Theta_R = \{m_\phi, \Lambda_\gamma, \beta_m, \text{parameters of } V(\phi), \text{screening}\}, \quad (74)$$

where Λ_γ sets the coupling to F^2 , and β_m sets the coupling to matter.

For phase geometry,

$$\Theta_P = \{\text{coefficients or kernels of } \Gamma_{\text{corr}}, \text{coarse-graining scale}\}. \quad (75)$$

One should begin with the lowest-dimensional form possible, for example a linear geometric response,

$$H_{\mu\nu}^{\text{corr}}(x) = \int d^4y K_{\mu\nu}{}^{\alpha\beta}(x, y) T_{\alpha\beta}(y), \quad (76)$$

subject to covariance, causality, and conservation constraints. Independent arbitrary functions for cosmology, galaxies, and the laboratory must not be allowed; otherwise the hypothesis loses falsifiability.

17.2 Stage 1: local constraints on a physical radion

This stage precedes the FAIR-MAST analysis. It determines whether any region of parameter space Θ_R remains in which a plasma effect could be measurable.

The following must be used jointly:

- a) the final MICROSCOPE test of the weak equivalence principle, which constrained composition-dependent acceleration at the order of 10^{-15} [19];
- b) torsion-balance and short-distance tests of the $1/r^2$ law, including measurements down to tens of micrometers [20];
- c) atomic-clock comparisons and spectroscopic limits on the temporal and gravitationally dependent variation of fundamental constants;
- d) searches for scalar-photon conversion and afterglow in strong magnetic fields, such as OSQAR and CHASE [21, 22];

- e) constraints on additional polarizations and anomalous propagation of gravitational waves.

The output should be a common likelihood,

$$\mathcal{L}_{\text{local}}(m_\phi, \Lambda_\gamma, \beta_m, \dots), \quad (77)$$

not a collection of unrelated numerical bounds. If the entire region capable of producing a detectable effect in a tokamak or in cosmology is already excluded, the simple H_R branch is rejected before any further fitting.

17.3 Stage 2: cosmological data

The cosmological analysis must separate the background, light propagation, and perturbation growth.

Core public data sets:

- DESI DR2 BAO measurements and cosmological chains [13];
- DES-SN5YR light curves and Hubble diagram [14], together with Pantheon+ [16];
- Planck 2018 temperature, polarization, and lensing likelihoods [17];
- public weak-lensing, structure-growth, cosmic-chronometer, and standard-siren data as they become available.

Background reconstruction.

First, without imposing a specific model, reconstruct

$$H(z), \quad D_M(z), \quad D_H(z), \quad D_L(z), \quad (78)$$

with the full covariance. To test the fixed point $a_\star \propto t$, calculate

$$t(z) = t_0 - \int_0^z \frac{dz'}{(1+z')H(z')}, \quad (79)$$

$$\Omega_{\text{rec}}(z) = \frac{a(z)}{a_\star[t(z)]} = \frac{t_0}{(1+z)t(z)}, \quad (80)$$

using the normalization $a_\star = t/t_0$. This is a diagnostic decomposition, not a proof: the microscopic model must subsequently predict $\Omega(t)$ independently.

Null and clock tests.

Any alternative to standard expansion must simultaneously reproduce

$$1+z, \quad D_L = (1+z)^2 D_A, \quad \Delta t_{\text{obs}} = (1+z)\Delta t_{\text{em}}, \quad (81)$$

where the last relation is tested by supernova light curves [15]. Additional tests are required for surface brightness, strong lensing, time delays, and future redshift drift.

Perturbations and geometric slip.

Lensing, clustering, and velocity data must be used to reconstruct

$$G_{\text{eff}}(k, z), \quad \eta_{\text{slip}}(k, z) = \frac{\Phi}{\Psi}, \quad f\sigma_8(z). \quad (82)$$

H_R must relate these functions to the same m_ϕ and the same couplings constrained by local experiments. H_P must explain them without an independent scalar pole.

Result criterion.

A cosmological deviation is informative only if one and the same parameterization consistently describes the background, time dilation, lensing, and structure growth. Fitting a single $D_L(z)$ diagram is not a test of the hypothesis.

17.4 Stage 3: galactic data

The first pass is performed on SPARC, after which the result must be tested on independent samples of rotation curves, gas profiles, and galaxy lensing.

Reconstruction of the baryonic model:

- a) one convention for Υ_{disk} and Υ_{bulge} ;
- b) preservation of the sign of the gas contribution where it occurs in the original decomposition;
- c) treatment of uncertainties in V_{obs} , distance, inclination, and correlated points within one galaxy;
- d) strict fixation of quality flags and preregistered cuts.

Model-independent RAR.

The mean function $\hat{m}(\log g_{\text{bar}})$ is constructed by a spline, Gaussian process, or hierarchical regression rather than by a MOND formula. Residuals are analyzed at the galaxy level so that thousands of radial points are not treated as independent.

Head-tail test and radial kernel.

Equations (68) and (69) are tested. Mandatory controls are

$$\{M_{\text{bar}}, V_f, R_d, R_{\text{eff}}, \Sigma_{\text{eff}}, f_{\text{gas}}, J_{\text{proxy}}, \text{morphology, data quality}\}. \quad (83)$$

Use robust regression, permutation tests within narrow mass and surface-density groups, leave-one-galaxy-out validation, and injection of synthetic signals.

Separation of the branches.

A physical radion must generate a universal response to energy and composition rather than a special dependence only on observed gas. The phase hypothesis permits a collective relation to the spatial organization of the baryonic state, but it must also reproduce lensing. Therefore, a positive head-tail effect must be compared with galaxy-galaxy lensing, satellite dynamics, and cluster dynamics.

17.5 Stage 4: FAIR-MAST

FAIR-MAST provides a public catalog of historical MAST campaigns, a JSON API for metadata, and diagnostic data in Zarr format through object storage [18]. This permits an archival test without operating a dedicated facility.

4.1. Availability audit

First construct a table of available pulses and channels:

$$\{\text{shot}, t_{\text{UTC}}, B_T, I_p, P_{\text{heat}}, \bar{n}_e, q_{95}, \text{plasma shape, gas, campaign, diagnostics}\}. \quad (84)$$

It is necessary to check separately whether sufficiently long series exist with reversals of B_T , I_p , or closely matched shots. If no such series exist, the sign test is replaced by regression on field amplitude and geometry.

4.2. Physical model for H_R

The minimal scalar correction is

$$(\square - m_\phi^2)\phi = \frac{1}{4}B_{F,\phi}F_{\mu\nu}F^{\mu\nu} + \beta_m T, \quad (85)$$

$$\nabla_\mu [B_F(\phi)F^{\mu\nu}] = J^\nu. \quad (86)$$

In a magnetically dominated regime, the source is

$$F_{\mu\nu}F^{\mu\nu} \simeq 2B^2. \quad (87)$$

Corrections to MHD, wave dispersion, and current diffusion must be derived from these equations before data analysis. The fit parameters must coincide with Θ_R from local and cosmological tests.

4.3. Observables

Candidates include:

- residuals of magnetic reconstruction and EFIT;
- loop voltage, current-profile evolution, and magnetic diffusion;
- frequencies and damping of MHD and Alfvén modes;
- energy-confinement time τ_E ;
- the radial electric field and plasma rotation;
- disruption probability at fixed standard parameters;
- vacuum coil-only and pre-plasma intervals as negative controls.

For matched shots with field-direction reversal, define

$$\Delta Y_{\text{odd}} = \frac{Y(+B_T) - Y(-B_T)}{2}, \quad \Delta Y_{\text{even}} = \frac{Y(+B_T) + Y(-B_T)}{2}. \quad (88)$$

Conventional drifts and the divertor configuration also change under reversal, so only a residual after the standard MHD model is of interest.

4.4. Terrestrial backgrounds

For each pulse, reconstruct the direction of the geomagnetic field and the orientation of the device. The cross term

$$B_{\text{tot}}^2 = B_{\text{device}}^2 + 2\mathbf{B}_{\text{device}} \cdot \mathbf{B}_{\oplus} + B_{\oplus}^2 \quad (89)$$

is already present in ordinary electrodynamics; only a residual after accounting for the full measured field can represent new physics. The gravitational difference in clock rate across a meter scale is of order 10^{-16} and is effectively inaccessible to plasma diagnostics by itself. A significant effect would require a nonstandard scalar or collective coupling.

A shot-level regression may take the form

$$Y_s = Y_{\text{MHD}}(\boldsymbol{\theta}_s) + \beta_F \overline{F^2}_s + \beta_{\oplus} \overline{\mathbf{B}_{\text{device}} \cdot \mathbf{B}_{\oplus}_s} + u_{\text{campaign}} + \epsilon_s. \quad (90)$$

Shuffled timestamps, fictitious orientations, empty channels, campaign splitting, and a blind analysis of the sign of $\beta_{\oplus}$ are required.

4.5. Interpretation

A resonant or frequency-dependent response consistent with (35) and local constraints would support H_R . Absence of such a signal constrains H_R but has little effect on H_P . Supporting H_P would require a reproducible collective residual without a free scalar pole and with the same response structure as that found in galactic or cosmological data.

17.6 Stage 5: joint inference and search for common structure

After the independent analyses, construct a joint likelihood,

$$\mathcal{L}_{\text{joint}} = \mathcal{L}_{\text{local}} \mathcal{L}_{\text{cosmo}} \mathcal{L}_{\text{gal}} \mathcal{L}_{\text{MAST}}. \quad (91)$$

The shape of the response is more important than individual best-fit parameters:

$$\chi_{\text{cosmo}}(k, z), \quad K_{\text{gal}}(r, r'), \quad \chi_{\text{MAST}}(\omega, k; B, n, T). \quad (92)$$

For H_R , the same scalar propagator with a common mass and couplings must appear in all three domains. For H_P , no universal one-particle pole is expected; instead, a scale-dependent collective susceptibility may occur, constrained by causality and gauge invariance.

Table 2: Decision logic.

Observed result	Most natural interpretation	Further action
Correlated WEP / α / G / GW / FAIR-MAST signatures of one scalar pole	Support for H_R	Joint fit of m_ϕ , couplings, and potential; test screening
No local scalar, but a robust galactic and cosmological nonlocal response with a common scale	Relative support for H_P	Derive a covariant Γ_{corr} and test lensing
SPARC correlations disappear after controlling for mass, brightness, and quality	Conventional astrophysics or endogeneity	Do not use them as evidence for new geometry

Observed result	Most natural interpretation	Further action
An anomaly appears only in FAIR-MAST and is not reproducible across campaigns and reversals	Diagnostic systematics or plasma physics	Negative controls and an independent facility
All new parameters are consistent with zero	No empirical support for H_R or H_P	Retain minimal GR+ED or seek different micro-dynamics

17.7 Stage 6: reproducibility and protection against overfitting

For every analysis, publish:

1. the exact source-data version and checksums;
2. a single code repository and execution environment;
3. preregistered cuts, observables, and null tests;
4. full covariance matrices and nuisance parameters;
5. posterior predictive checks and injection of synthetic signals;
6. separation of exploratory and confirmatory samples;
7. results of null checks even when no positive effect is found.

The output of the program should be not merely a yes/no answer, but machine-readable constraints on response functions. Even negative results will narrow the class of admissible microscopic models.

18 Falsification Criteria

18.1 Common core

The common causal-phase hypothesis must be rejected or substantially modified if:

1. a causal structure with local $U(1)$ transports has no appropriate Lorentzian continuum limit;
2. the discrete phase functional does not reproduce the Maxwell action;

3. causal order and the volume measure do not consistently reconstruct the same metric scale;
4. joint coarse-graining violates gauge invariance, diffeomorphism invariance, or causality;
5. the same effective functional cannot simultaneously describe the dynamics of massive bodies, lensing, and light propagation;
6. the claimed galactic relations are fully explained by endogeneity of definitions, systematics, or ordinary astrophysics;
7. cosmological observables cannot be reproduced by one consistent metric and one parameter set.

18.2 Physical-radion branch H_R

The simple H_R version is rejected if the parameter region required for a cosmological or plasma effect is inconsistent with MICROSCOPE, clocks, short-distance gravity, scalar-photon conversion searches, and gravitational waves. Separate screening mechanisms are admissible only when supplied by an explicit model with new testable consequences.

18.3 Phase-space branch H_P

The H_P branch loses empirical content if Γ_{corr} can be selected independently for each data set, or if coarse-graining leaves only renormalizations of G , e , and Λ without new linked predictions. Detection of a universal free scalar pole moves the model into the H_R class or another physical scalar theory.

18.4 Fixed point $a_\star \propto t$

This additional hypothesis is rejected as physically useful if the reconstructed deformation $\Omega(t)$ is merely an arbitrary reparameterization of the data, is not derived from microdynamics, has no monotonic relation to an independent statistical functional, or requires violation of BAO, CMB, supernova time dilation, and lensing.

19 Final Formulation

The fundamental structure is assumed to be locally finite and causally ordered. Causal order determines the null structure, volume information determines the metric scale, and $U(1)$ holonomies determine the phase connection. Proper time is the length of a timelike worldline. Light is an excitation of the connection and a null probe of the full effective metric. Two competing interpretations of the phase layer remain. In H_R , the layer is a physical five-dimensional geometry and contains a radion; in H_P , it is an internal phase space without a free scalar mode. The minimal limit coincides with Einstein–Maxwell theory.

A limiting geometry $g_{\mu\nu}^*$ with $a_* \propto t$ may be investigated, but the observed metric must be an independently derived nonequilibrium deformation, not a rewriting of the known $a(t)$. Entropic “synchronization” is understood as a decrease of relative entropy toward a fixed point.

The choice among H_R , H_P , and the null model must be made jointly using local scalar tests, cosmology, galactic dynamics and lensing, and FAIR-MAST. Only a unified response structure with common parameters counts as a positive result. Absence of such a structure is grounds for rejecting both branches.

(93)

Thus, the current version is a research program:

causal order + volume measure + $U(1)$ holonomies $\xrightarrow{\text{coarse-graining}} (g_{\mu\nu}, A_\mu, \Gamma_{\text{corr}})$ $\xrightarrow{\text{data}}$ physical radion, phase space, or refutation.

(94)

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